Dmitrii Nikolaev, Ph.D.

Senior ML Engineer | LLM Inference | Distributed GPU Training

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PROFILE

Machine learning engineer with a Ph.D. in Applied Mathematics and over seven years of industry experience. At AMD, I own enterprise customers' LLM inference and training performance on AMD and NVIDIA GPUs, from initial benchmarks to production. On recent projects, my changes let the same GPUs handle 55% more concurrent requests and made training 75% faster. Before that, I built diffusion-based avatar generation services, LLM evaluation and retrieval systems at Silo AI, and earlier taught ML at university, where I supervised eight students on applied ML projects and mentored them through problem formulation, experimentation, and implementation.

SKILLS

- **LLM inference:** vLLM, SGLang, KV-cache management, continuous batching, quantization, memory allocation tuning and profiling, multi-GPU serving, latency and throughput benchmarking.
- **Training and post-training:** PyTorch, DeepSpeed, Megatron-LM, data/tensor/pipeline/sequence parallelism, LoRA fine-tuning, multi-LoRA merging, evaluation pipelines.
- **GPU performance:** ROCm, CUDA, roofline analysis, profiling, GEMM tuning, new hardware acceptance benchmarking, C++20 kernels debugging and development for NPU.
- **RAG and LLM evaluation:** PostgreSQL-backed hybrid retrieval, vector and keyword search, LLM re-ranking, expert-judgment evaluation sets, AI-code detection pipelines.
- **Agentic benchmarking:** self-hosted vLLM endpoints, agents that modify code, allocate cloud GPUs, run parameter sweeps, and assemble reports; Claude Code workflows.
- **Cloud and MLOps delivery:** AWS, Azure AI Foundry, managed Azure AI endpoints, Kubernetes, Helm, Docker, Slurm, Prometheus, Grafana, CI/CD, MLflow.

EXPERIENCE

AMD **Helsinki, Finland**

Semiconductors and AI compute, joined via the Silo AI acquisition

Staff AI Engineer *Aug 2024 – Present*

- **Agentic benchmarking.** Built an internal AMD harness for intrusive testing and knowledge organization: open-weight LLMs served through self-hosted vLLM endpoints drive agents that modify code, allocate cloud GPUs, run experiments, and assemble reports.
- **LLM inference.** Ran long throughput and latency tests on a production vLLM and SGLang deployment for a top-tier cloud infrastructure provider, serving 40000 tokens per second on an 8-GPU cluster. Unblocked concurrent inference of the 8-bit-quantized MiniMax 2.5 model, traced a slowdown to a critical indexing bug in AITER, and resolved memory contention through KV-cache shuffling and quantization settings. As a result, the same GPUs handled 55% more concurrent requests.
- **Distributed training.** For a global electronics manufacturer, operated DeepSpeed workloads and built a benchmark library that pinpoints bottlenecks across data, tensor, pipeline, and sequence parallelism. Roofline analysis showed the pipeline setup did not fit the hardware; after retuning GEMM kernels and the parallel layout, training ran **75%** faster.
- **Hardware evaluation.** Benchmarked new GPU generations on ROCm and CUDA and turned the results into deployment instructions for customer engineering teams.
- **Customer technical lead.** Worked directly with customer engineering teams to understand their technical goals and constraints, scoped and prioritized optimization work, defined success metrics, presented findings and trade-offs, and aligned on next steps.
- **Edge inference.** Ported PyTorch inference to next-generation FPGA accelerators and AMD XDNA spatial AI-engine arrays, writing tiled C++20 operators that fit within on-chip memory. Inference ran **1.3x** faster for CNNs and **2.6x** faster for Transformers than on the integrated GPU, compared against a reference implementation.

Silo AI **Helsinki, Finland**
AI lab for enterprise clients, acquired by AMD in 2024
Senior AI Engineer *Nov 2023 – Aug 2024*

- **LLM evaluation.** Built a system that distinguishes AI-written code from human-written code, using LLM-based generation and evaluation pipelines. The first deployed version reached about **90%** precision at **80%** recall and cut manual review by about **60%**.
- **RAG Search.** Delivered patent search for a global telecom company, combining PostgreSQL-backed hybrid retrieval with LLM re-ranking. Prior art searches went from hours to minutes, with quality measured against expert relevance judgments.
- **Fine-tuning.** Owned the avatar-generation service from concept to production. Fine-tuned diffusion models with LoRA and combined subject-specific and style-specific adapters to preserve user identity while creating professional profile photos in new visual styles, without retraining the full model.
- **Predictive pricing.** Delivered a LightGBM pricing model on Azure AI Foundry and served it through managed Azure AI endpoints for an industrial access-systems client.

Skoltech, AI R&D Lab **Moscow, Russia**
Research university lab building ML systems for industrial partners
Software / Research Engineer *Jul 2019 – Apr 2023*

- Designed and owned a multi-modal esports analytics platform funded by Riot Games, combining telemetry, video, text, and biometric data into real-time predictions, dashboards, and anomaly alerts for live events.
- Built a reinforcement learning pipeline with self-play and opponent modeling for a game AI client. It matched an expert rule-based system and replaced hand-written rules.
- Integrated tensor-storage data lakes into distributed training pipelines; tuned PyTorch data loaders and dataset migration, removing the input bottleneck for GPU training jobs.

FusionSoft **Lipetsk, Russia**
Software development studio
Software Engineer *Dec 2017 – Mar 2019*

- Built GPU job orchestration for a Kubernetes-native MLOps platform for distributed training and synthetic-data generation: dynamic per-job cluster provisioning from YAML manifests, reproducible model-delivery pipelines, and GitHub Actions CI/CD.
- Designed and developed a high-load enterprise analytics platform replacing legacy SAP modules; reduced analytics query time from days to seconds for an international client.

Lipetsk State Technical University **Lipetsk, Russia**
Technical university and applied mathematics department
Assistant / Associate Professor *Feb 2010 – Dec 2017*

- Taught machine learning, optimization, numerical methods, and probability theory; supervised 8 graduate students and delivered applied ML/control prototypes, including robot localization and control.

EDUCATION

Voronezh State Technical University | *Ph.D. in Applied Mathematics* **2010–2014**
Thesis: numerical methods for modeling and control of multi-agent systems — developed neural-network-driven coordination and control of cooperating agents.

Lipetsk State Technical University | *Specialist in Applied Mathematics* **2005–2010**
Five-year degree, equivalent to B.Sc. + M.Sc. Thesis: computer vision and algebraic methods for autonomous robot control — built neural-network control of a robotic hand under physics-based constraints.

ADDITIONAL INFORMATION

Location	Helsinki, Finland (EU Blue Card; mobile within EU; open to remote and relocation)
Languages	Russian (native), English (full working proficiency), Finnish and German (conversational)